

FCR

FCR is a cancer drug treatment. It is used to treat chronic lymphocytic leukaemia (CLL) and some types of non-Hodgkin lymphoma (NHL).

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What is FCR?

FCR is a combination of cancer drugs. It is used to treat [chronic lymphocytic leukaemia \(CLL\)](#) and some types of [non-Hodgkin lymphoma \(NHL\)](#).

It is best to read this information with our general information about [cancer drug treatments](#) and the [type of cancer](#) you have.

FCR is named after the initials of the drugs used:

- F – [fludarabine](#)
- C – [cyclophosphamide](#)
- R – [rituximab](#).

Fludarabine and cyclophosphamide are types of chemotherapy. Rituximab is a type of targeted therapy called a [monoclonal antibody](#).

Your cancer team will talk to you about this treatment and its possible side effects before you agree ([consent](#)) to have treatment.

More information about this treatment

This information is correct at time of publishing. But sometimes the types of cancer this treatment is used for, or treatment side effects, may change between revision dates.

You can talk to your cancer team if you want more detailed information about this treatment. Or visit the [electronic Medicines Compendium \(eMC\)](#) website, which has patient information leaflets (PIL) for individual drugs.

How FCR is given

You will have FCR in a chemotherapy day unit or clinic as an outpatient. Or you may have it during a stay in hospital.

During your course of treatment, you will meet someone from your cancer team, such as a:

- cancer doctor
- chemotherapy nurse or a specialist nurse
- specialist pharmacist.

This is who we mean when we mention doctor, nurse or pharmacist in this information.

Before or on the day of each treatment, you will have a blood test. This is to check that it is safe for you to have treatment.

You will meet with a doctor, nurse or pharmacist before you have treatment. They will talk to you about your blood results and ask how you have been feeling. If your blood results are okay, the pharmacy team will prepare your cancer drugs.

Your nurse will usually give you anti-sickness drugs before the cancer drugs. The cancer drugs can be given in different ways. You may have:

- rituximab as a drip into a vein (intravenous) - after the first treatment, you might have it as an injection under the skin (subcutaneous)
- fludarabine and cyclophosphamide as drips or injections into a vein or as tablets.

Your nurse will explain what to expect. Any drugs into a vein will be given through 1 of the following:

- a [cannula](#) – a short, thin tube the nurse puts into a vein in your arm or hand
- a [central line](#) – a fine tube that goes under the skin of your chest and into a vein close by
- a [PICC line](#) – a fine tube that is put into a vein in your arm and goes up into a vein in your chest
- an [implantable port](#) (portacath) – a disc that is put under the skin on your chest or arm and goes into a vein in your chest.

Your course of chemotherapy

You have a course of several cycles of treatment over a few months. Each cycle of FCR usually takes 28 days (4 weeks). You have the cancer drugs on the first days of the cycle - for example, on days 1, 2 and 3. Then you have a break with no treatment until the start of the next cycle.

Your cancer team will discuss your treatment plan with you. They will explain what drugs you will have on what days. They may give you a copy of the plan to take home.

Taking fludarabine and cyclophosphamide tablets

If your treatment involves tablets, your cancer team will give you these to take home. They may give you tablets of different strengths. Always take them exactly as explained. This is important to make sure they work as well as possible for you.

You can take fludarabine and cyclophosphamide with or without food. Make sure you:

- swallow them whole with a glass of water
- do not chew, break or crush them
- take them at the same time on the days you take them.

If you forget to take the tablets, take the missed dose as soon as possible within the same day. If a full day has passed, do not take a double dose. Contact the hospital on the 24-hour number you have been given for advice.

Other things to remember about your tablets:

- Wash your hands after taking them.
- Other people should avoid direct contact with them.
- Keep them in the original package and at room temperature, away from moisture, heat and direct sunlight.
- Keep them safe, where children cannot see or reach them.
- If you are sick (vomit) just after taking them, do not take an extra dose. Take your next dose at the usual time.

- Do not throw away unused tablets. Return them to your cancer team at the hospital.

Your cancer team may also give you anti-sickness drugs and other medicines to take home. Take all your capsules or tablets exactly as they tell you to.

About side effects

We explain the most common side effects of this treatment here. We also include some that are less common.

You may get some of the side effects we mention, but you are unlikely to get all of them. And you may have some side effects, including rarer ones, that we have not listed here.

Always tell your doctor, nurse or pharmacist about any side effects you have. They can give you:

- drugs to help control some side effects
- advice about managing side effects.

It is important to take any drugs exactly as explained. This means they will be more likely to work for you.

Serious and life-threatening side effects

Some cancer treatments can cause serious side effects. Sometimes, these may be life-threatening. Your doctor, nurse or pharmacist can explain the risk of these side effects to you.

Contact the hospital

Your doctor, nurse or pharmacist will give you 24-hour contact numbers for the hospital. If you feel unwell or need advice, you can call at any time of the day or night. Save these numbers in your phone or keep them somewhere safe.

Side effects while treatment is being given

Some people may have side effects while they are having a cancer drug or shortly after they have it:

Allergic reaction

Some people have an allergic reaction to rituximab or cyclophosphamide while they are having it. A reaction is most likely with the first rituximab infusion, so it is given slowly over a few hours. Before treatment, you will be given medicines to help prevent or reduce any reaction.

Signs of a reaction can include:

- feeling hot or flushed
- a skin rash
- itching

- shivering
- feeling dizzy
- a headache
- feeling breathless
- swelling of your face or mouth
- pain in your back, tummy or chest.

Your nurse will check you for signs of a reaction during your infusion. If you feel unwell or have any of these signs, tell them straight away. If you do have a reaction, it can be treated quickly.

Sometimes a reaction can happen a few hours after treatment. If you develop any signs or feel unwell after you get home, contact the hospital straight away on the 24-hour number.

Blood pressure

Your blood pressure may fall when you are having rituximab. If you usually take medicine to lower your blood pressure, your doctor, nurse or pharmacist may ask you not to take it for 12 hours before having rituximab. Sometimes, rituximab can make your blood pressure go up. Your nurse will check your blood pressure regularly.

Flushes and blocked nose

Some people who have cyclophosphamide have hot flushes or feel like their nose is blocked while the drug is being given. You may also have a strange taste in your mouth. These side effects do not last for long. But if you notice this, ask the nurse to slow down the drip. This will reduce these symptoms.

Tumour pain

During the rituximab infusion, some people have mild pain in the parts of the body where they have cancer. Painkillers can help with this.

Very common side effects

These side effects happen to 10 or more people in every 100 people (10% or more) who have this treatment.

Risk of infection

This treatment can reduce the number of white blood cells in your blood. These cells fight infection. If your white blood cell count is low, you may be more likely to get an [infection](#). A low white blood cell count is called neutropenia.

An infection can be very serious when the number of white blood cells is low. It is important to get any infection treated as soon as possible. If you have any of the following symptoms, contact the hospital straight away on the 24-hour number:

- a temperature above 37.5°C

- a temperature below 36°C
- you feel unwell, even with a normal temperature
- you have symptoms of an infection.

Symptoms of an infection include:

- feeling shivery and shaking
- a sore throat
- a cough
- breathlessness
- diarrhoea
- needing to pass urine (pee) often, or discomfort when you pass urine.

It is important to follow any specific advice your cancer treatment team gives you.

Your white blood cell count will usually return to normal before your next treatment. You will have a blood test before having more treatment. If your white blood cell count is low, your doctor may delay your treatment for a short time, until your cell count increases.

Bruising and bleeding

This treatment can reduce the number of platelets in your blood. Platelets are cells that help the blood to clot.

If the number of platelets is low, you may bruise or bleed easily. You may have:

- nosebleeds
- bleeding gums
- heavy periods
- blood in your urine (pee) or stools (poo)
- tiny red, brown or purple spots that may look like a rash – these spots can be harder to see if you have black or brown skin.

If you have any unexplained bruising or bleeding, contact the hospital straight away on the 24-hour number. You may need a drip to give you extra platelets. This is called a platelet transfusion.

Anaemia (low number of red blood cells)

This treatment can reduce the number of red blood cells in your blood. Red blood cells carry oxygen around the body. If the number of red blood cells is low, this is called anaemia. You may feel:

- very low in energy
- breathless
- dizzy and light-headed.

If you have these symptoms, contact the hospital straight away on the 24-hour number. You may need treatment for anaemia. If you are very anaemic, you may need a drip to give you extra red blood cells. This is called a blood transfusion.

Feeling sick

Your doctor, nurse or pharmacist will prescribe anti-sickness drugs to help prevent or control sickness. Take the drugs exactly as they tell you to, even if you do not feel sick. It is easier to prevent sickness than to treat it after it has started.

[If you feel sick](#), take small sips of fluid often and eat small amounts regularly. It is important to drink enough fluids. If you continue to feel sick, or if you are sick (vomit) 1 to 2 times in 24 hours, contact the hospital on the 24-hour number as soon as possible. They will give you advice. They may change your anti-sickness treatment. Let them know if you still feel sick.

Bladder irritation

Cyclophosphamide may irritate your bladder and cause discomfort when you pass urine. Drink plenty of fluids – at least 2 litres (3½ pints) during chemotherapy and in the 24 hours after having it. It is also important to empty your bladder regularly and to try to pass urine as soon as you feel the need to go.

Tell your nurse or doctor straight away if you feel any discomfort or stinging when you pass urine or if you notice any blood in your urine.

Diarrhoea

This treatment may cause [diarrhoea](#). Diarrhoea means passing more stools (poo) than is normal for you, or having watery or loose stools. You may also have stomach cramps. If you have a stoma, it may be more active than usual.

If you are passing loose stools 3 or more times a day and this is not normal for you, contact the hospital as soon as possible on the 24-hour number. Follow the advice they give you about:

- taking anti-diarrhoea medicines
- drinking enough fluids to keep you hydrated and to replace lost salts and minerals
- any changes to your diet that might help.

They might also ask you for a specimen of your stool to check for infection.

Feeling tired

[Feeling tired](#) is a common side effect of this treatment. It is often worse towards the end of treatment and for some weeks after it ends. Try to pace yourself and plan your day so you have time to rest. Gentle exercise, like short walks, can help you feel less tired.

If you feel sleepy, do not drive or use machinery.

Hair loss

Your hair will get thinner. Or you may lose all the hair from your head. You may also lose your eyelashes and eyebrows, as well as other body hair. Hair loss usually starts after your first or second treatment.

If you want to cover up hair loss, there are different ways you can do this. Your nurse will give you information about [coping with hair loss](#).

Remember to [protect your skin from the sun](#). Use suncream with a sun protection factor (SPF) of at least 30 on your scalp. Or cover up with a hat or scarf.

Hair loss is almost always temporary. Your hair will usually grow back after treatment ends.

Skin changes

Chemotherapy can affect your skin. If your skin feels dry, try using soap-free cleansers and unperfumed moisturising cream every day.

This treatment can also:

- cause a rash, which may be itchy
- make your skin darker in some areas
- make any area treated with radiotherapy become red or sore. If you have white skin the area will become red and if you have black or brown skin the area might become darker.
- make you more sensitive to the sun
- prevent wounds from healing.

Your skin may burn more easily during treatment and for several months after. Use a sun cream of at least SPF 30. SPF stands for sun protection factor. Cover up with clothing and wear a hat.

Always tell your doctor, nurse or pharmacist about any [skin changes](#). They can give you advice or prescribe creams or medicines to help. Changes to your skin are usually temporary and improve when treatment ends.

Other side effects

These side effects happen to less than 10 in 100 people (less than 10%) who have this treatment. Some of them are much rarer than this but they are still important to know about. Rare means a side effect that happens to less than 1 in 1,000 people (less than 0.1%).

Raised blood sugar levels

This treatment may raise your blood sugar levels and increase the risk of diabetes.

You will have regular blood tests to check this.

Contact the hospital on the 24-hour number if you have these symptoms during treatment or after it ends:

- feeling thirsty all the time
- needing to pass urine (pee) more often than usual
- feeling more tired than usual.
- losing weight without trying.

If you have diabetes, your blood sugar levels may be higher than usual. You may need advice and support from your GP or diabetes team about managing this.

Difficulty sleeping (insomnia)

Sometimes this treatment can affect your sleep. Tell your doctor, nurse or pharmacist if cancer treatment makes it [difficult to sleep](#).

Hearing changes

This treatment may cause hearing changes, including hearing loss. You may have ringing in the ears. This is called tinnitus. You may also become unable to hear some high-pitched sounds. Hearing changes may get better after this treatment ends. But this does not always happen. If you notice any changes in your hearing, tell your doctor, nurse or pharmacist.

Numb or tingling hands or feet (peripheral neuropathy)

This treatment may affect the nerves in your fingers and toes. This can cause numbness, tingling or pain in your hands or feet. This is called [peripheral neuropathy](#). You might find it hard to do fiddly tasks such as fastening buttons or tying shoelaces.

If you have these symptoms, always tell your doctor, nurse or pharmacist. They sometimes need to change the drug or the dose of the drug. The symptoms usually improve slowly after treatment ends. But for some people they continue and are a long-term side effect of treatment.

Fluid build-up

This treatment can cause a build-up of fluid in the body. This will slowly get better after treatment ends. Contact the hospital on the 24-hour number if you:

- are gaining weight
- have swelling in your face, legs or ankles.

They can give you advice and treatment to help.

Sore mouth and throat

This treatment may cause a [sore mouth and throat](#). You may also get mouth ulcers. This can make you more likely to get a mouth or throat infection. Use a soft toothbrush to clean your teeth or dentures in the morning, at night and after meals.

Contact the hospital straight away on the 24-hour number, if:

- a sore mouth or throat affects how much you can drink or eat
- your mouth, tongue, throat or lips have any blisters, ulcers or white patches.

They can give you advice, and mouthwash or medicines to help with the pain or to treat any infection. Follow their advice and make sure you:

- drink plenty of fluids
- avoid alcohol and tobacco
- avoid food or drinks that irritate your mouth and throat.

Effects on the eyes

Your eyes may become watery and feel sore. Your doctor, nurse or pharmacist can give you advice and eye drops to help with this.

Contact the hospital on the 24-hour number if:

- your eyes get red and inflamed (conjunctivitis)
- you have pain in your eyes
- you notice any change in your vision.

Tumour lysis syndrome (TLS)

Some people are at risk of developing a condition called tumour lysis syndrome (TLS). The risk is highest at the beginning of treatment. TLS happens when treatment makes large numbers of cancer cells die and break down quickly. This releases lots of waste products into the blood and can affect the kidneys and heart.

TLS can be prevented or treated. You will have regular blood tests to check for TLS. You may have:

- extra fluids through a drip
- medicines such as allopurinol tablets or [rasburicase](#) through a drip.

Drinking at least 2 litres (3½ pints) of fluid a day will also help.

Effects on the lungs

This treatment can cause changes to the lungs. Tell your doctor, nurse or pharmacist if you develop:

- a cough that does not go away
- wheezing
- breathlessness.

You should also tell them if any existing breathing problems get worse. You may have tests to check your lungs.

Effects on the heart

This treatment can affect how the heart works. You may have [tests](#) to check how well your heart is working. These may be done before, during and after treatment.

If the treatment is causing [heart problems](#), your doctor may change the type of treatment you are having.

Contact the hospital straight away on the 24-hour number if you have any of these symptoms during or after treatment:

- breathlessness
- dizziness
- changes to your heartbeat (palpitations)
- swollen feet and ankles.

Other conditions can cause these symptoms, but it is important to get them checked by a doctor.

Always call **999** if you have:

- chest pain, pressure, heaviness, tightness or squeezing across the chest
- difficulty breathing.

Serious skin changes

Rarely, this treatment can cause a serious skin reaction that needs to be treated immediately in hospital.

Do not take any more of this treatment and contact the hospital straight away on the 24-hour number if you have any of these symptoms:

- a skin rash that is spreading
- blistering or peeling skin
- flu-like symptoms, such as a high temperature and joint pain
- sores on your lips or in your mouth.

Effects on the nervous system

Rarely, this treatment can affect the brain and nerves. These effects may be mild at first but can become serious if not treated. Symptoms may include:

- weakness, numbness or tingling in your arms, legs or face
- a headache that does not get better
- seeing double or having drooping eyelids
- difficulty turning or bending your neck
- drowsiness or confusion
- problems with moving or speaking
- fits (seizures).

If you have any of these symptoms during treatment or after it ends, it is important to either:

- contact the hospital straight away on the 24-hour number
- go to the hospital straight away
- call **999**.

Do not drive yourself to hospital.

Second cancer

This treatment can increase the risk of developing a second cancer years later. This is rare. The benefits of treatment usually far outweigh this risk. Your doctor, nurse or pharmacist can give you more information.

Other information

Blood clot risk

Cancer and some cancer treatments can increase the risk of a [blood clot](#). Contact the hospital straight away on the 24-hour number if you have any of these symptoms during or after treatment:

- throbbing pain or swelling in a leg or arm
- reddening of the skin in the area – if you have black or brown skin, this can be harder to notice, but the skin might become darker
- suddenly feeling breathless or coughing.

Always call **999** if you have:

- chest pain
- difficulty breathing.

A blood clot is serious, but it can be treated with drugs called anticoagulants. These thin the blood. Your doctor, nurse or pharmacist can give you more information about preventing and treating blood clots.

Hepatitis B reactivation

If you have had hepatitis B (a liver infection) in the past, rituximab can make it active again. Your doctor or nurse will talk to you about this and test you for hepatitis B. If you have active hepatitis B, you will not have treatment with rituximab.

Other medicines

Some medicines can affect how this treatment works or be harmful while you are having it. Always tell your cancer doctor, nurse or pharmacist about any drugs you are taking or planning to take, such as:

- medicines you have been prescribed
- medicines you buy in a shop, pharmacy or online
- vitamins or supplements
- [herbal drugs and complementary or homeopathic therapies](#)
- recreational drugs – for example, cannabis.

Alcohol

Drinking alcohol while taking cyclophosphamide may make some side effects worse, such as feeling sick. You should follow sensible drinking guidelines while having this treatment.

Lactose

This treatment may contain lactose. If you have a lactose allergy or intolerance, ask your doctor, nurse or pharmacist for more information.

Irradiated blood and platelets

After this treatment, if you need blood or platelet transfusions, they should be treated with radiation (irradiated) before you have them. This helps prevent a rare but serious reaction that can be caused by white blood cells in the donated blood.

Irradiated blood is not radioactive and will not harm you or anyone you come into contact with.

Some people must have any blood or platelet transfusions irradiated for the rest of their lives. If you need this, your doctor will give you more information. They will give you a medical alert card to carry in case you need blood in an emergency or at another hospital.

Vaccinations

Cancer doctors usually recommend that people with cancer have vaccinations for flu and [coronavirus \(covid\)](#). They may also recommend other vaccines, such as Shingrix[®] for shingles. These all help reduce your risk of serious illness from these infections. Most people can have these vaccines, including people with weak immune systems.

You should not have live vaccines if your immune system is weak. This includes if you are having or recently had chemotherapy, radiotherapy or other cancer treatments that affect your immune system. Live vaccines can make you unwell because they contain a very weak version of the illness they protect you against. There are several live vaccines, including the yellow fever vaccine.

It is important to ask your doctor, nurse or pharmacist for advice about having vaccinations. They can explain what vaccines are right for you and when it is best to have them.

Contraception

Your doctor, nurse or pharmacist will advise you not to [get pregnant](#) or [make someone pregnant](#) while having this treatment and for some time afterwards. The drugs may harm a developing baby. It is important to use contraception to prevent pregnancy. Follow their advice about:

- what types of contraception to use
- how long after treatment you should continue to use contraception.

Breastfeeding

You are advised not to [breastfeed](#) while having this treatment, or for some time after treatment ends. This is because the drugs could be passed to the baby through breast milk.

Your doctor, nurse or pharmacist can give you more information.

Fertility

Some cancer drugs can affect whether you can [get pregnant](#) or [make someone pregnant](#). If you are worried about this, it is important to talk with your doctor before you start treatment.

Sex

It is possible that small amounts of chemotherapy may be passed on through vaginal fluids or semen. If you have sex during this treatment, your cancer team will usually advise using condoms or a dental dam to protect your partner.

Medical and dental treatment

If you need medical treatment for any reason other than cancer, always tell the healthcare professional that you are having cancer treatment. Give them the contact details for your cancer doctor or cancer team so they can ask for advice.

If you have appointments with a dentist, always tell them you are having cancer treatment. Talk to your cancer team before you have any dental treatment.

About our information

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer.

References



Visit the [electronic Medicines Compendium \(eMC\)](#) to download a Patient Information Leaflet (PIL) for more detailed information. The leaflet lists all known side effects.

[Cancer Information Development team](#)

Author

Date reviewed

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Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
-

use illustrations to explain text

- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

How we can help

How to print a PDF from this website

Instructions on how you can make a PDF version of this page using the print function on your browser.

Clinical Information Nurse Specialists

Our Cancer Information Nurse Specialists are dedicated cancer nurses available to talk to on our Macmillan Cancer Support Line.



Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help